

Notes of Optimization

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LECTURE 1 OF 03/03/2026

We will give a quick summary on basic notions of topology, calculus 1 & 2.

DEFINITION 1 (Norm/scalar product). Given $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ the Euclidean norm (or ℓ^2 norm) is defined as

$$\|x\| = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}.$$

For $x, y \in \mathbb{R}^n$ the *scalar product* is given by

$$\langle x, y \rangle = x \cdot y = x^\top y = \sum_{i=1}^n x_i y_i.$$

Moreover, the scalar product induces a norm. Indeed, one can verify that

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

DEFINITION 2 (Open ball). Given $x \in \mathbb{R}^n$, $r > 0$ we call the *open ball of radius r centered in x* the set

$$B(x, r) = \{y \in \mathbb{R}^n : \|y - x\| < r\}.$$

DEFINITION 3 (Limit of a sequence). Let $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}^n$, $\bar{x} \in \mathbb{R}^n$. We say that x_n converges to $\bar{x}$, and write $x_n \rightarrow \bar{x}$ if for every $\varepsilon > 0$ there exists an $N > 0$ such that for every $k > N$, $x_k \in B(\bar{x}, \varepsilon)$, i.e. for k big enough, $\|\bar{x} - x_k\| < \varepsilon$, and we say that

$$\lim_{k \rightarrow +\infty} x_k = \bar{x}.$$

Remark 4. The limit in $\mathbb{R}^n$ is the pointwise limit, so $x_k \rightarrow \bar{x}$ if and only if $(x_k)_j \rightarrow (\bar{x})_j$ for any $1 \leq j \leq n$.

Example 5. Consider the sequence $\{x_k\}_{k \in \mathbb{N}}$ defined as

$$x_k = \left(\frac{1}{k}, \frac{1}{k} \sin(k) \right) \rightarrow (0, 0)$$

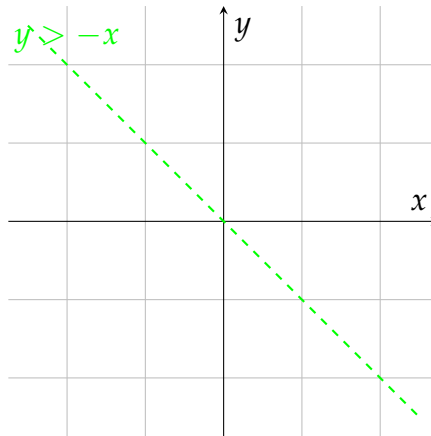


Figure 1.1: Ball of radius r centered in x on $\mathbb{R}^2$.

as $k \rightarrow +\infty$.

DEFINITION 6 (Open/closed set). Let $A \subseteq \mathbb{R}^n$. We say that A is *open* if for every $x \in A$ there exists an $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$. We say that A is *closed* if $\mathbb{R}^n \setminus A$ is open.

Example 7. Consider the set

$$A = \left\{ (x_1, x_2) \in \mathbb{R}^2 : |x_1| < 1, |x_2| < 1 \right\}.$$

We can easily verify that A is open. Observe that a set can be neither open nor closed. The sets $\mathbb{R}^n$ and $\emptyset$ are both open and closed at the same time.

DEFINITION 8 (Neighbour). We say that, for a given $x \in \mathbb{R}^n$, a subset $A \subseteq \mathbb{R}^n$ is a *neighbour* of x if A is an open set containing x .

DEFINITION 9 (Interior, exterior, boundary, closure). Let $A \subseteq \mathbb{R}^n$, $x \in \mathbb{R}^n$. We say that x is:

- an *interior point* for A if $\exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$. We denote the set of all interior points of A with A° ;
- A *boundary point* for A if $\forall \varepsilon > 0$, $B(x, \varepsilon) \cap A \neq \emptyset$ and $B(x, \varepsilon) \cap A^c \neq \emptyset$. We denote the set of all the boundary points with ∂A ;
- an *exterior point* for A if $\exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A^c$.

We denote the *closure* of A as $\bar{A} = A \cup \partial A$.

PROPOSITION 10. Let $\{A_k\}_{k \in \mathbb{N}}$ a sequence of open sets. Then:

i) the countable union of open sets is open, i.e.

$$\bigcup_{k \in \mathbb{N}} A_k \text{ is open;}$$

ii) the finite intersection of open sets is open, i.e. for $N \in \mathbb{N}$,

$$\bigcap_{i=1}^N A_i \text{ is open.}$$

Proof. We prove i) and ii) separately.

i): set $E = \bigcup_{k=1}^{+\infty} A_k$. Given $x \in E$ then there exists $m > 0$ such that $x \in A_m$. Since A_m is open, there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A_m \subseteq E$, so we found an open ball containing x fully contained in E . Since the choice of x is arbitrary, this must hold for every $x \in E$, hence E is open.

ii): set $E = \bigcap_{k=1}^N A_k$. Let $x \in E$, then $x \in A_k$ for every $1 \leq k \leq N$, hence there exists a sequence $\varepsilon_k > 0$ such that $B(x, \varepsilon_k) \subseteq A_k$ for every $k < N$. Consider $\bar{\varepsilon} = \frac{1}{2} \min_k \varepsilon_k > 0$. Then $B(x, \bar{\varepsilon}) \subseteq B(x, \varepsilon_k)$ for every k , hence $B(x, \bar{\varepsilon}) \subseteq E$. Again, since the choice of x is arbitrary, this must hold true for every $x \in E$, hence E is open. ■

PROPOSITION 11 (Characterization of closed sets). Let $C \subseteq \mathbb{R}^n$. Then C is closed $\iff$ for every sequence $\{x_k\}_{k \in \mathbb{N}} \subseteq C$ such that $x_k \rightarrow \bar{x}$, then $\bar{x} \in C$.

Proof. We prove $\implies$ and $\impliedby$.

" $\implies$ ": let $C \subseteq \mathbb{R}^n$ be closed. Consider $\{x_k\}_{k \in \mathbb{N}}$ such that $x_k \rightarrow \bar{x} \in \mathbb{R}^n$, we want to show that $\bar{x} \in C$. Assume, by contradiction, that $\bar{x} \notin C \implies \bar{x} \in \mathbb{R}^n \setminus C$, which is an open set, so there exists $\varepsilon > 0$ such that $B(\bar{x}, \varepsilon) \subseteq \mathbb{R}^n \setminus C \implies B(\bar{x}, \varepsilon) \cap C = \emptyset$, but $x_k \rightarrow \bar{x}$, so, for this ε , there must be an $N > 0$ such that $x_k \in B(\bar{x}, \varepsilon)$ for $k > N$, which is in contradiction with the fact that the sequence takes values in C .

" $\Leftarrow$ ": let C be a set. Consider $\{x_k\}_{k \in \mathbb{N}} \subseteq C$ such that $x_k \rightarrow \bar{x} \in C$. Assume, by contradiction, C not closed. Then, for every $\varepsilon > 0$, there exists $z \in \mathbb{R}^n \setminus C$ such that $B(z, \varepsilon) \cap (\mathbb{R}^n \setminus C) = \emptyset$ and $B(z, \varepsilon) \cap C \neq \emptyset$. So we can find a sequence $\{x_k\}_{k \in \mathbb{N}}$ with values in C such that $x_k \rightarrow z$ by fixing a sequence $\{\varepsilon_k\}_{k \in \mathbb{N}}$ such that $\varepsilon_k \searrow 0$, so $x_k \in B(z, \varepsilon_k) \cap C$, $x_k \rightarrow z$, so, by assumption $z \in C$, which is in contradiction with $z \in \mathbb{R}^n \setminus C$. ■

DEFINITION 12 (Bounded/finite set). A set $A \subseteq \mathbb{R}^n$ is *bounded* if there exists $R > 0$ such that $A \subseteq B(0, R)$.

The set A is *finite* if $A = \{x_1, \dots, x_m\} \subseteq \mathbb{R}^n$.

DEFINITION 13 (Continuity). Let $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\bar{x} \in E$. We say that f is *continuous* in $\bar{x}$ if for every sequence $\{x_k\}_{k \in \mathbb{N}}$ such that $x_k \rightarrow \bar{x}$ in $\mathbb{R}^n$, then $f(x_k) \rightarrow f(\bar{x})$ in $\mathbb{R}^m$.

DEFINITION 14 (Cluster/isolated point). Let $E \subseteq \mathbb{R}^n$, $\bar{x} \in \mathbb{R}^n$. We say that $\bar{x}$ is a *cluster point* for E if there exists a sequence $\{x_k\}_{k \in \mathbb{N}}$ such that, for every $\bar{x} \neq x_k \rightarrow \bar{x}$. We denote the set of all cluster points with $\text{Acc}(E)$.

A point $\bar{x}$ is *isolated* in E if $x \in E \setminus \text{Acc}(E)$.

DEFINITION 15 (Limit). Let $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\bar{x} \in \text{Acc}(E)$. Fixed $\ell \in \mathbb{R}^m$, we say that

$$\lim_{x \rightarrow \bar{x}} f(x) = \ell$$

if for every sequence $\{x_k\}_{k \in \mathbb{N}}$, $x_k \rightarrow \bar{x}$, $x_k \neq \bar{x}$ for every k , we have $f(x_k) \rightarrow \ell$. Notice that we allow $\bar{x}$ to not necessarily be in E .

Example 16. Consider the function $f : \{0\} \cup [1, 2] \subset \mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x$. Then it has no meaning to speak about $\lim_{x \rightarrow 0} f(x)$, as $0 \notin \text{Acc}(\{0\} \cup [1, 2])$, while f is continuous at 0, since the only sequence converging to 0 in $\{0\} \cup [1, 2]$ is the sequence which is identically 0.

Remark 17. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous, then:

- $\{x : f(x) < c\}$ is *open* $\forall c \in \mathbb{R}$;
- $\{x : f(x) \leq c\}$ is *closed* $\forall c \in \mathbb{R}$;
- $\{x : f(x) = c\}$ is *closed* $\forall c \in \mathbb{R}$.

CHAPTER 2

LECTURE 2 OF 04/03/2026

Our main focus from now on is to solve the problem

$$\inf_{x \in A} f(x)$$

for some $A \subseteq \mathbb{R}^n$ and some $f : \mathbb{R}^n \rightarrow \mathbb{R}$. We will restrict ourselves only to certain functions and sets.

DEFINITION 1 (Infimum/supremum). Let $E \subseteq \mathbb{R}$. We denote with $\inf E =$ largest $z \in \mathbb{R}$ such that $z \leq x$ for every $x \in E$. By convention we set $\inf \emptyset = +\infty$.

Similarly, we denote with $\sup E$ the smallest $z \in \mathbb{R}$ such that $z \geq x$ for every $x \in E$, with the convention $\sup \emptyset = -\infty$. More formally,

$$\begin{aligned} z = \inf E & \quad \text{if } z \leq x \forall x \in E \text{ and } \forall \varepsilon > 0 \exists y \in E \text{ s.t. } y \leq z + \varepsilon \\ z = \sup E & \quad \text{if } z \geq x \forall x \in E \text{ and } \forall \varepsilon > 0 \exists y \in E \text{ s.t. } y \geq z - \varepsilon \end{aligned}$$

Remark 2. The inf and sup aren't necessarily inside the set. If they are, we speak of min and max.

Axiom 1 — Completeness of $\mathbb{R}$

Every bounded sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}$ admits a supremum in $\mathbb{R}$.

THEOREM 3 (Bolzano-Weierstrass). Any bounded sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}$ admits a cluster point, i.e. there exists $\bar{x} \in \mathbb{R}$ and a subsequence $\{x_{n_j}\}_{j \in \mathbb{N}}$ such that $x_{n_j} \rightarrow \bar{x}$ as $j \rightarrow +\infty$.

Proof. Since $\{x_n\}_n$ is bounded, there exists $R > 0$ such that $x_n \in [-R, R]$ for every $n \in \mathbb{N}$. Set $I_1 = [-R, R]$, select x_{n_1} a value in I_1 . Divide I_1 in half and select the one subinterval containing infinitely many x_n s, call it $I_2 = [a_2, b_2]$ (here $a_2 = -R, b_2 = 0$ or $a_2 = 0, b_2 = R$). We have $I_2 \subset I_1$, $|I_2| = |I_1|/2$, and choose x_{n_2} a value in I_2 . Repeat this process to cut and select. We end up, at the k -th step, with a sequence of subintervals $I_1, I_2, \dots, I_k$ and 3 subsequences: $x_{n_1}, x_{n_2}, \dots, x_{n_k}$

the sequence of selected values, $a_1, a_2, \dots, a_k$ the sequence of left boundaries and $b_1, b_2, \dots, b_k$ the sequence of right boundaries.

We observe that $a_{j+1} \geq a_j$ and $b_{j+1} \leq b_j$ for all $1 \leq j \leq k$. Moreover, $a_j \leq R$ and $b_j \geq -R$ for all j , so, by the axiom of completeness of $\mathbb{R}$,

- there exists $\bar{a} \in \mathbb{R}$ such that $a_k \rightarrow \bar{a} = \sup_k a_k$
- there exists $\bar{b} \in \mathbb{R}$ such that $b_k \rightarrow \bar{b} = \inf_k b_k$

We observe that $\bar{a} = \bar{b}$. Indeed, $|\bar{b} - \bar{a}| = |\bar{b} - b_k + b_k - a_k + a_k - \bar{a}| \leq |\bar{b} - b_k| + |\bar{a} - a_k| + |b_k - a_k|$. The terms $|\bar{b} - b_k|$ and $|\bar{a} - a_k|$ goes to 0 as $k \rightarrow +\infty$ for what we observed before. Moreover, $|b_k - a_k| = 2R/2^{k-1} \rightarrow 0$ as $k \rightarrow +\infty$.

Set $\bar{x} = \bar{a} (= \bar{b})$. Then the subsequence $\{x_{n_j}\}_j$ is converging to $\bar{x}$: indeed $|x_{n_j} - \bar{x}| = |x_{n_j} - a_j + a_j - \bar{x}| \leq |a_{n_j} - a_j| + |a_j - \bar{x}|$.

Since $a_j \rightarrow \bar{x} = \bar{a}$, the term $|a_j - \bar{x}| \rightarrow 0$. Notice that, by construction, $x_{n_j} \in I_j$, so $|x_{n_j} - a_j| \leq |I_j| \rightarrow 0$.

■

DEFINITION 4. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. We say that $\bar{x} \in \mathbb{R}^n$ is a *local maximum* for f if there exists a neighbourhood U of $\bar{x}$ such that $f(x) \leq f(\bar{x})$ for every $x \in U$. A *local strict maximum* has the strict inequality.

Similarly, $\bar{x}$ is *local minimum* for f if $\exists U$ neighbourhood of $\bar{x}$ such that $f(x) \geq f(\bar{x})$ for all $x \in U$.

If there exists $\bar{x}$ such that $f(x) = \inf_{x \in A} f(x)$, the $\bar{x}$ is a *global minimum* over A .

DEFINITION 5. We denote with $\arg_{x \in A} \min f(x)$ the set of minimum points for f over A , i.e

$$\arg_{x \in A} \min f(x) = \left\{ \bar{x} \in \mathbb{R}^n : f(\bar{x}) = \inf_{x \in A} f(x) \right\}.$$

THEOREM 6 (Weierstrass). Let $A \subseteq \mathbb{R}^n$ be a compact set, $f : A \rightarrow \mathbb{R}$ a continuous function. Then there exists $x_m, x_M \in A$ such that

$$f(x_m) = \min_{x \in A} f(x)$$

$$f(x_M) = \max_{x \in A} f(x).$$

Proof. We prove the statement for the minimum point.

Let $\ell = \inf_{x \in A} f(x)$. We can always find a sequence $\{x_n\}_n \subseteq A$ such that $f(x_n) \rightarrow \ell$ by the continuity of f . Since A is bounded, $\{x_n\}_n$ is bounded as well. By Bolzano-Weierstrass, there exists a subsequence $\{x_{n_k}\}_k$ converging to $\bar{x} \in A$ by closeness of A . By continuity, in particular f is continuous at $\bar{x}$, ■

Remark 7. For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we define the *gradient* of f as

$$\nabla f(c) = \left(\frac{\partial f}{\partial x_1}(c), \dots, \frac{\partial f}{\partial x_n}(c) \right)$$

the vector of partial derivatives, defined as

$$\frac{\partial f}{\partial x_i}(c) = \lim_{h \rightarrow 0} \frac{f(c + he_i) - f(c)}{h}$$

whenever this limit exists finite, and where e_i is the i -th vector of the canonical base of $\mathbb{R}^n$. Furthermore, we say that f is *differentiable* at c if

$$f(x) = f(c) + \langle \nabla f(c), x - c \rangle + \mathcal{O}(\|x - c\|^2)$$

for $x \rightarrow c$.

We now set our focus on convex analysis.

DEFINITION 8 (Convex set). Let $C \subseteq \mathbb{R}^n$. We say that C is *convex* if, for any $x, y \in C$, and for any $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha)y \in C$.

PROPOSITION 9. a) $\bigcap_{i \in I} C_i$ is convex if C_i is convex for any $i \in I$ finite or countable;

b) the algebraic sum $C_1 + C_2 = \{x + y : x \in C_1, y \in C_2\}$ is convex if C_1 and C_2 are convex;

c) the set $\lambda C = \{\lambda x : x \in C\}$ is convex if C is convex for any $\lambda \in \mathbb{R}$;

d) $\overline{C}, C^\circ$ are convex if C is convex;

e) if a function $f : C \rightarrow \mathbb{R}$ is affine, then the set $f(C) = \{y : y = f(x), x \in C\}$ is convex if C is convex.

Proof. We prove all the propositions separately.

a): suppose C_i convex and let $C = \bigcap_{i \in I} C_i$. Let $x, y \in C, \alpha \in [0, 1]$. Since $x, y \in C$, in particular $x, y \in C_i$ for all $i \in I$. But then also $\alpha x + (1 - \alpha)y \in C_i$ for all $i \in I$, hence C is convex.

b): let $x = x_1 + x_2 \in C_1 + C_2, x_1 \in C_1$ and $x_2 \in C_2$, take $y \in C_1 + C_2$ of the same form.

$$\alpha x + (1 - \alpha)y = \underbrace{\alpha x_1 + (1 - \alpha)y_1}_{\in C_1} + \underbrace{\alpha x_2 + (1 - \alpha)y_2}_{\in C_2} \in C.$$

c): trivial.

d): consider $\{x_k\}_k \subseteq C$ and $\{y_k\}_k \subseteq C$ such that $x_k \rightarrow x$ and $y_k \rightarrow y$. Then the sequence $\{\alpha x_k + (1 - \alpha)y_k\}_k \subseteq C$. But then also $\alpha x + (1 - \alpha)y \in C$ as limit of sequences valued in C .

Let now $x, y \in C^\circ$. Then there exists $r > 0$ such that $B(x, r) \subseteq C$ and $B(y, r) \subseteq C$. Consider $\alpha x + (1 - \alpha)y$. Our claim is: $B(\alpha x + (1 - \alpha)y, r) \subseteq C$. Indeed, any $z \in B(\alpha x + (1 - \alpha)y, r)$ is of the form $z = \alpha x + (1 - \alpha)y + w$ with $\|w\| < r$, so $\alpha(x + w) + (1 - \alpha)(y + w) \in C$ by convexity of C , hence the claim. ■

LECTURE 3 OF 10/03/2026

Example 1 (Convex sets). The following sets are convex in $\mathbb{R}^n$:

- **hyperplanes**: $\{x \in \mathbb{R}^n : a^\top x = b\}$ for $a \in \mathbb{R}^n, b \in \mathbb{R}$;
- **halfspaces**: $\{x \in \mathbb{R}^n : a^\top x \leq b\}$ for $a \in \mathbb{R}^n, b \in \mathbb{R}$;
- **polyhedra**: $\{x \in \mathbb{R}^n : a_i^\top x \leq b_i\}$ for $a_i \in \mathbb{R}^n, b_i \in \mathbb{R}$ for $i = 1, \dots, m$ (they are intersection of half spaces).

Half cones are convex as well.

DEFINITION 2 (Convex/concave function). Let $C \subseteq \mathbb{R}^n$ be convex. Consider $f : C \rightarrow \mathbb{R}$. We say that f is *convex* if the following holds:

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y) \quad \forall x, y \in C, \forall \alpha \in [0, 1].$$

We say that f is *concave* if $-f$ is convex.

Example 3. Affine functions are convex. Indeed, let $f(x) = a^\top x + b, a \in \mathbb{R}^n$.
Indeed,

$$\begin{aligned} f(\alpha x + (1 - \alpha)y) &= \langle a, \alpha x + (1 - \alpha)y \rangle + b(1 - \alpha + \alpha) \\ &= \alpha \langle a, x \rangle + \alpha b + (1 - \alpha) \langle a, y \rangle + (1 - \alpha)b \\ &= \alpha f(x) + (1 - \alpha)f(y). \end{aligned}$$

Example 4. The norm in $\mathbb{R}^n$ is convex, as an easy consequence of triangular inequality.

We can observe that f convex implies that the set $V_\gamma = \{x \in \mathbb{R}^n : f(x) \leq \gamma\}$ is convex for all $\gamma \in \mathbb{R}$. Indeed, for every $x, y \in V_\gamma$ we have $f(x) \leq \gamma$ and $f(y) \leq \gamma$. We see that $f(\alpha x + (1 - \alpha)y) \in V - \gamma$. Indeed, by convexity of f we have $f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y) = \alpha\gamma + (1 - \alpha)\gamma = \gamma$.

In general, the converse is not true, i.e. convex level sets *does not imply* f convex.
From now on, $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$, where $\bar{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}$.

DEFINITION 5 (Epigraph). Let $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$. We define the set

$$\text{epi}(f) = \left\{ (x, w) \in \mathbb{R}^{n+1} : x \in X, w \in \mathbb{R}, f(x) \leq w \right\}.$$

An intuitive definition could be "the set of points above the graph of f ".

DEFINITION 6 (Effective domain). Given $f : X \subseteq \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ we define the *effective domain* of f as the set

$$\text{dom}(f) = \{x \in X : f(x) < +\infty\} = \{x \in X : \exists w \in \mathbb{R} \text{ s.t. } (x, w) \in \text{epi}(f)\}.$$

DEFINITION 7 (Proper function). A function $f : X \subseteq \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ is said *proper* if $f(x) < +\infty$ and $f(x) > -\infty$ for all $x \in X$. In other words, $\text{epi}(f)$ does not contain any vertical line.

DEFINITION 8 (Extension of convex functions). Let $C \subseteq \mathbb{R}^n$ be a convex set. A function $f : C \rightarrow \overline{\mathbb{R}}$ is *convex* if $\text{epi}(f)$ is convex in $\mathbb{R}^{n+1}$.

Remark 9. If $f : C \rightarrow \mathbb{R}$ the definitions coincides.

Proof. Assume $\text{epi}(f)$ convex in $\mathbb{R}^{n+1}$, then the points $(x, f(x)), (y, f(y)) \in \text{epi}(f)$, but also $\alpha(x, f(x)) + (1 - \alpha)(y, f(y)) \in \text{epi}(f)$. This implies $(\alpha x + (1 - \alpha)y, \alpha f(x) + (1 - \alpha)f(y)) \in \text{epi}(f) \implies f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$. ■

DEFINITION 10 (Convex indicator function). Given $A \subseteq \mathbb{R}^n$, we define the *convex indicator function* of the set A as

$$\delta_A(x) = \begin{cases} 0 & \text{if } x \in A \\ +\infty & \text{otherwise} \end{cases}$$

This is coherent with the infimum operation, as

$$\inf_{x \in A} f(x) = \inf_{x \in \mathbb{R}^n} f(x) + \delta_A(x).$$

DEFINITION 11 (Liminf). Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}, c \in \text{Acc}(X)$. We say that

$$\ell = \liminf_{x \rightarrow c} f(x)$$

if $\ell = \inf_{x_k} \lim_{k \rightarrow +\infty} f(x_k)$ where $\{x_k\}_k$ is a sequence in X , $\{x_k\}_k \rightarrow c$, $x_k \neq c$ for all k and $\lim_{k \rightarrow \infty} f(x_k)$ exists finite.

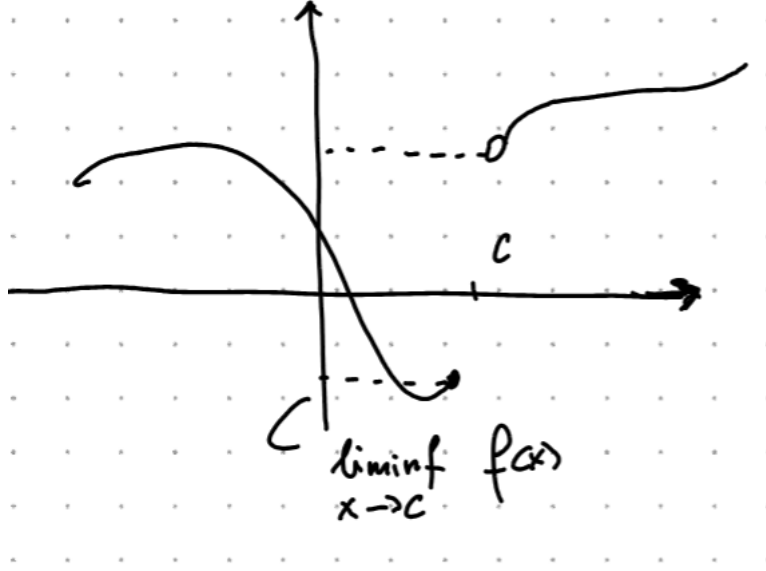


Figure 3.1: The $\liminf$ of f at $x = 1$ is 1, even if $f(1) = 0.5$.

DEFINITION 12 (Lower semicontinuity). Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. We say that f is *lower semicontinuous* (lsc for short) at $c \in X$ if

$$f(c) \leq \liminf_{x \rightarrow c} f(x).$$

THEOREM 13 (Weierstrass). *The Weierstrass theorem still allows a minimum point for a lsc function.*

PROPOSITION 14. Let $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$. The following are equivalent:

- i) The level sets $V_\gamma = \{x \in \mathbb{R}^n : f(x) \leq \gamma\}$ are closed for all $\gamma \in \mathbb{R}$;
- ii) f is lsc;
- iii) $\text{epi}(f)$ is closed in $\mathbb{R}^{n+1}$.

Proof. We prove $i) \implies ii) \implies iii) \implies i)$.

$i) \implies ii)$: suppose V_γ closed for all γ . Assume by contradiction f not lsc, so there exists $c \in \mathbb{R}^n$ s.t. $f(c) > \liminf_{x \rightarrow c} f(x)$. This implies that there exists a sequence $\{x_k\}_k$, $x_k \rightarrow c$ and a $\gamma \in \mathbb{R}$ s.t. $f(c) > \gamma \geq \lim_{k \rightarrow +\infty} \inf f(x_k)$. For k big enough, we have $f(x_k) \leq \gamma$, so $x_k \in V_\gamma$. By closeness of V_γ , it must be $c \in V_\gamma \implies f(c) \leq \gamma$, which is in contradiction with $f(c) > \gamma$.

ii) $\implies$ iii): assume f lsc, we want to show that any sequence $\{(x_k, w_k)\}_k \subseteq \text{epi}(f)$ converging to $(\bar{x}, \bar{w})$ must satisfy $(\bar{x}, \bar{w}) \in \text{epi}(f)$. Assume $x_k \rightarrow \bar{x}$, $w_k \rightarrow \bar{w}$. Then $f(x_k) \leq w_k$ for all k . Since $x_k \rightarrow \bar{x}$ and f is lsc, it must be that $f(\bar{x}) \leq \liminf_{k \rightarrow +\infty} f(x_k) \leq \liminf_{k \rightarrow +\infty} w_k = \bar{w}$, hence $(\bar{x}, \bar{w}) \in \text{epi}(f)$.

iii $\implies$ i): assume $\text{epi}(f)$ closed. Let $\gamma \in \mathbb{R}$, $V_\gamma = \{x : f(x) \leq \gamma\}$. Consider the sequence $\{x_k\}_k \subseteq V_\gamma$ s.t $x_k \rightarrow \bar{x}$. We want to show $\bar{x} \in V_\gamma$. The points $(x_k, \gamma) \in \text{epi}(f)$ for all k , so, by closeness of $\text{epi}(f)$, $(\bar{x}, \gamma) \in \text{epi}(f)$. This implies $f(\bar{x}) \leq \gamma$, hence $\bar{x} \in V_\gamma$. ■

PROPOSITION 15. Let $f : \mathbb{R}^m \rightarrow (-\infty, +\infty]$, $A \in \mathbb{R}^{m \times n}$, $F : \mathbb{R}^n \rightarrow (-\infty, +\infty]$, $x \mapsto f(Ax)$. If f is convex, then F is convex. If f is closed, then F is closed.

Proof. Assume f convex. Then

$$F(\alpha x + (1 - \alpha)y) = f(\alpha Ax + (1 - \alpha)Ay) \leq \alpha f(Ax) + (1 - \alpha)f(Ay).$$

Assume f closed, so $\text{epi}(f)$ is closed and f is lsc. Take a sequence $x_k \rightarrow \bar{x}$. Then $\liminf_k F(x_k) = \liminf_k f(Ax_k)$. Let $y_k = Ax_k$, so $y_k \rightarrow A\bar{x}$ by continuity of the linear map Ax . This implies $\liminf_k f(Ax_k) \geq f(A\bar{x}) = F(\bar{x})$. ■

PROPOSITION 16. Let $\{f_i\}_{i=1}^m$ be a family of convex and closed function s.t. $f_i : \mathbb{R}^n \rightarrow (-\infty, +\infty]$, and $\gamma_1, \dots, \gamma_m > 0$. Then the function defined by $F(x) = \sum_{i=1}^m \gamma_i f_i(x)$ is closed and convex.

PROPOSITION 17. Let $f_i : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ for $i \in I$ any countable index set. Let $f(x) = \sup_i f_i(x)$. If f_i is closed convex for all $i \in I$ then f is closed convex.

Proof. Let $(x, w) \in \text{epi}(f)$, then $f(x) \leq w$, so $f_i(x) \leq w$ for all i . This implies $(x, w) \in \text{epi}(f_i)$ for all i , hence $\text{epi}(f) = \bigcap_{i \in I} \text{epi}(f_i)$, which is convex as countable intersection of convex sets. The same argument holds for closeness. ■

LECTURE 4 OF 11/03/2026

We ask ourselves what happens if f is convex and we add some regularity hypothesis. We first remark that if f is differentiable at c , near this point, the function looks like an hyperplane. The intuition is: function must be always above its tangent hyperplane.

PROPOSITION 1. *Let $C \subseteq \mathbb{R}^n$ be convex, $A \subseteq \mathbb{R}^n$ open s.t. $C \subseteq A$. Let $f : A \rightarrow \mathbb{R}$ convex and differentiable. Then*

- i) f is convex on $C \iff f(z) \geq f(x) + \langle \nabla f(x), z - x \rangle$ for all $x, z \in C$
- ii) f is strictly convex over $C \iff f(z) > f(x) + \langle \nabla f(x), z - x \rangle$ for all $x, z \in C$ s.t. $x \neq z$

Proof. Let f be convex on C . Let $x, z \in C$. Observe that

$$\lim_{h \rightarrow 0^+} \frac{f(x + h(z - x)) - f(x)}{h} = \frac{\partial}{\partial h} f(x + h(z - x))|_{h=0} = \nabla f(x) \cdot (z - x).$$

Then

$$\begin{aligned} f(x) + \langle \nabla f(x), z - x \rangle &= f(x) + \lim_{h \rightarrow 0^+} \frac{f(x + h(z - x)) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{hf(x) - f(x) + f(hz + (1 - h)x)}{h} \\ &\leq \lim_{h \rightarrow 0} \frac{-(1 - h)f(x) + hf(z) - (1 - h)f(x)}{x} \\ &= \lim_{h \rightarrow 0} f(z) = f(z). \end{aligned}$$

Let now $\alpha \in [0, 1]$, $x, y \in C$, $z = \alpha x + (1 - \alpha)y$.

$$\begin{aligned} f(x) &\geq f(z) + \langle \nabla f(z), x - z \rangle \cdot \alpha \\ f(y) &\geq f(z) + \langle \nabla f(z), y - z \rangle \cdot (1 - \alpha) \end{aligned}$$

Take the sum of the above quantities, then

$$\alpha f(x) + (1 - \alpha)f(y) \geq \alpha f(z) + (1 - \alpha)f(z) + \alpha \langle \nabla f(z), x - z \rangle + (1 - \alpha) \langle \nabla f(z), y - z \rangle$$

hence, by linearity of the scalar product,

$$\alpha f(x) + (1 - \alpha)f(y) \geq f(z) + \langle \nabla f(z), \alpha x + (1 - \alpha)y - z \rangle.$$

Hence, f is convex.

For the strict convexity, we repeat the same process with the strict inequality. ■

If at some point x^* we have $\langle \nabla f(x^*), z - x^* \rangle \geq 0$ for all z , then x^* is a minimum point. This is also a necessary condition: assume x^* minimum point for f over C and, by contradiction, that there exists $z \in C$ s.t. $\langle \nabla f(x^*), z - x^* \rangle < 0$, but then

$$\frac{\partial}{\partial h} f(x^* + h(z - x^*))|_{h=0} = \langle \nabla f(x^*), z - x^* \rangle < 0$$

so f decreases in the direction $z - x^*$, which contradicts the fact that x^* is a minimum.

We summarize this fact in the following proposition, which proof what we just did.

PROPOSITION 2. Let $C \subseteq \mathbb{R}^n$ be convex, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable on a convex open set $A \subseteq C$. Then a point $x^* \in C$ is a minimum for f if and only if

$$\langle \nabla f(x^*), z - x^* \rangle \geq 0 \quad \forall z \in C.$$

THEOREM 3 (Projection theorem). Let C be a nonempty convex closed subset of $\mathbb{R}^n$, $z \in \mathbb{R}^n$. Then there exists a unique vector $x^* \in C$ (the projection of z on C) that minimizes the function $f(x) = \|x - z\|$ over C .

Furthermore, the point x^* is the projection of z onto C if and only if

$$\langle z - x^*, x - x^* \rangle \leq 0 \quad \forall x \in C.$$

Proof. Optimizing the function $f(x) = \|z - x\|$ is the same as optimizing the function $f(x) = \frac{1}{2}\|z - x\|^2$. Then f is now differentiable, and $\nabla f(x) = z - x$.

$$\begin{aligned} x^* \text{ is minimum} &\iff \langle \nabla f(x^*), x - x^* \rangle \geq 0 \quad \forall x \in C \\ &\iff \langle -x^* + z, x - x^* \rangle \leq 0. \end{aligned}$$

Does such an x^* exist? We observe that

$$\min_{x \in C} f(x) = \min_{x \in C \cap \{y: \|y-z\| \leq \|z-w\|\}, w \in C} f(x).$$

We notice that we are optimizing a continuous function on a closed set, as we are optimizing on $C \cap \overline{B(z, \|z-w\|)}$, so, by Weierstrass there exists a minimum. This is also unique, indeed, let x_1^*, x_2^* be two minima, then

$$\begin{aligned} \langle z - x_1^*, x_2^* - x_1^* \rangle &\leq 0 \quad \forall z \in C \\ \langle z - x_2^*, x_1^* - x_2^* \rangle &\leq 0 \quad \forall z \in C \end{aligned}$$

Taking the sum and using the linearity of the scalar product, we obtain

$$\langle x_1^*, x_2^*, z - x_2^* - z + x_1^* \rangle \leq 0 \implies \|x_1^* - x_2^*\| \leq 0 \implies x_1^* = x_2^*.$$

■

Convex hulls

The idea is to "convexify" sets that are not convex. Let X be a set. The *convex hull* of X $\text{conv}(X)$ = intersection of all convex sets containing X . A convex combination of elements of X is a vector in the form $\sum_{i=1}^m \alpha_i x_i$, where x_i are points in X , $\alpha_i \geq 0$ and $\sum_i \alpha_i = 1$.

Remark 4. If X is convex, any convex combination of elements of X belongs to X .

Remark 5.

$$\text{conv}(X) = \left\{ z = \sum_{i=1}^m \alpha_i x_i : \alpha_i \geq 0, x_i \in X, \sum_i \alpha_i = 1, \forall m > 0 \right\}.$$

Remark 6. If $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear transformation, then $\text{conv}(Ax) = A \text{conv}(X)$.

Remark 7. $\text{conv}\left(\sum_i X_i\right) = \sum_i \text{conv}(X_i)$.

DEFINITION 8 (Affine hull). Let $X \subseteq \mathbb{R}^n$. The *affine hull* of X , denoted with $\text{aff}(X)$ is the intersection of all affine sets containing X .

Recall that M is an affine set if M is in the form $M = \bar{x} + S$, where S is a linear subspace of $\mathbb{R}^n$.

Notice that $\text{conv}(X) \subseteq \text{aff}(X)$, and intersection of affine spaces is still an affine space, so $\text{aff}(X)$ is an affine space, so $\text{aff} X = \bar{x} + S$ for some $\bar{x} \in \mathbb{R}^n$ and S subspace. We define the dimension of $\text{aff}(X)$ as the dimension of S . S is also called the parallel subspace of the affine space.

Remark 9. $\text{aff}(X) = \text{aff}(\text{conv}(X)) = \text{aff}(\text{cl}(X))$ where $\text{cl}(X)$ denotes the closure of X .

DEFINITION 10 (Cone generated by X). Let $X \subseteq \mathbb{R}^n$. We define the *cone generated by X* the set

$$\text{cone}(X) = \left\{ \sum_{i=1}^m \alpha_i x_i : \alpha_i \geq 0, x_i \in X, m = 0 \right\}.$$

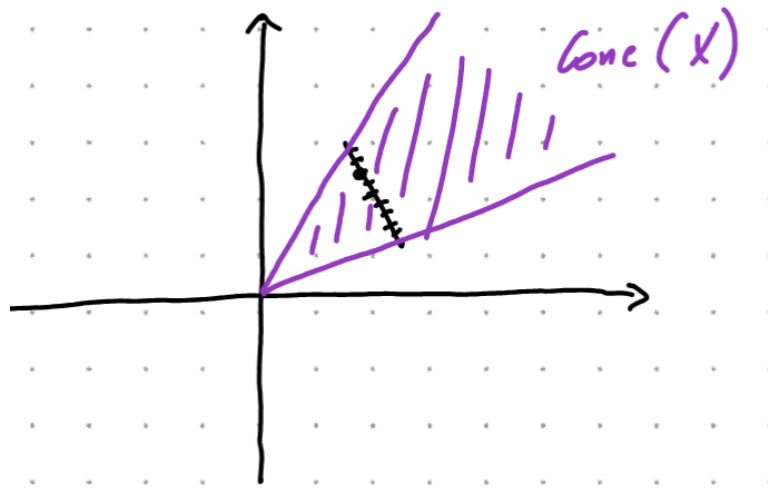


Figure 4.1: X (black) and cone generated by X (purple)

LECTURE 5 OF 18/03/2026

THEOREM 1 (Charateodory). *Let $X \subseteq \mathbb{R}^n$ be nonempty. Then*

- a) every nonzero vector from $\text{cone}(X)$ can be represented as a positive combination of linearly independent vectors in X ;*
- b) every vector from $\text{conv}(X)$ can be represented as a convex combination of no more than $n + 1$ vectors in X .*

Proof. We prove a) and b) separately.

a): Let $y \in \text{cone}(X) \implies y = \sum_{i=1}^m \alpha_i x_i$, where $\alpha_i > 0$, $x_i \in X$. We can assume m to be the smallest index such that this representation holds true. We have to prove that $x_1, \dots, x_m$ to be linearly independent. By contraddiction, assume them dependent, so there exists $\lambda_1, \dots, \lambda_m \in \mathbb{R}$ such that $\sum_{i=1}^m \lambda_i x_i = 0$.

Without loss of generality, assume at least one $\lambda_i > 0$. Define $\bar{\gamma} = \max \{ \gamma : \alpha_i - \gamma \lambda_i \geq 0 \} < +\infty$. Observe that we can write $y = \sum_i \alpha_i x_i + \underbrace{\bar{\gamma} \sum_i \lambda_i x_i}_{=0} = \sum_{i=1}^m (\alpha_i - \bar{\gamma} \lambda_i) x_i$. Then

at least one of the last coefficients is zero, which is in contraddiction with the minimality of m .

b): Consider the set $Y = \{(y, 1) : y \in X\} \subseteq \mathbb{R}^{n+1}$. Let $x \in \text{conv}(X) \implies x = \sum_{i=1}^m \gamma_i x_i$, $\sum_i \gamma_i = 1$. The point $(x, 1) \in \text{cone}(Y) \implies \sum_i \gamma_i (x_i, 1) = (x, 1)$.

By point a), $(x, 1) = \sum_{i=1}^m \alpha_i (\tilde{x}_i, 1)$, with $(\tilde{x}_i, 1)$ linearly independent in Y . Hence

$x = \sum_{i=1}^m \alpha_i \tilde{x}_i$ with $\sum_i \alpha_i = 1$. From the fact that $(\tilde{x}_i, 1)$ are independent in Y , we conclude that $m \leq n + 1$.

■

PROPOSITION 2. Let $X \subseteq \mathbb{R}^n$ be compact (closed and bounded). Then $\text{conv}(X)$ is compact.

Proof. It is easy to show that $\text{conv}(X)$ is bounded. Since X is bounded, $\exists R > 0$ such that $\|x\| \leq R$ for any $x \in X$. Then $x \in \text{conv}(X) \iff x = \sum_i \gamma_i x_i \implies \|x\| \leq \sum_i \gamma_i \|x_i\| \leq \sum_i \gamma_i R = R$.

Closeness: Take a sequence $\{x_k\}_k \subseteq \text{conv}(X)$ and assume $x_k \rightarrow \bar{x}$ as $k \rightarrow +\infty$. We want to show that $\bar{x} \in \text{conv}(X)$. Each x^k can be written as

$$x_k = \sum_{i=1}^m \alpha_i^k x_i^k$$

by the Carathéodory's theorem, with $\sum_i \alpha_i = 1, \alpha_i \geq 0, x_i \in X$. Consider the sequence $\{(\alpha_1^k, \dots, \alpha_{n+1}^k, x_1^k, \dots, x_{n+1}^k)\} \subseteq \mathbb{R}^{n+1+n(n+1)} = \mathbb{R}^{(n+1)^2}$. This is a bounded sequence and converges (up to subsequences by BW) to $(\alpha_1, \dots, \alpha_{n+1}, x_1, \dots, x_{n+1})$. Observe that, since $\sum_i \alpha_i^k = 1 \implies \sum_i \alpha_i = 1, \alpha_i \geq 0$. Observe furthermore that all $x_i \in X$ by closeness of X , so $\sum_i \alpha_i^k x_i^k \rightarrow \sum_i \alpha_i x_i$, hence $\bar{x} = \sum_{i=1}^{n+1} \alpha_i x_i$, hence $\bar{x} \in \text{conv}(X)$. ■

Relative interiors

If we consider a convex set in $\mathbb{R}^n$ which lives on an m -dimensional variety, with $m < n$ (as in 5.1), we get that the interior points of that convex sets forms the emptyset.

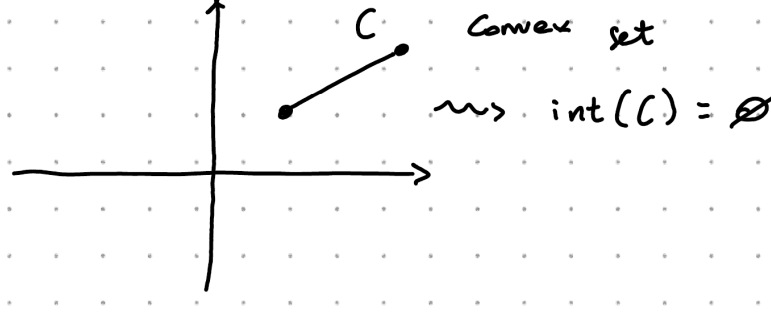


Figure 5.1: Example of emptiness of interior

DEFINITION 3 (Relative interior point). Let $C \subseteq \mathbb{R}^n$ be a convex set, $c \in C$. We say that x is a *relative interior point* if there exists an $r > 0$ such that $B(x, r) \cap \text{aff}(C) \subseteq C$.

The *relative interior* is the set of all the relative interior points, and we denote it with $\text{ri}(C)$.

Furthermore we say that C is *relatively open* if $C = \text{ri}(C)$.

PROPOSITION 4 (Line-segment principle (LSP)). Let C be a nonempty convex set. If $x \in \text{ri}(C)$ and $\bar{x} \in \text{cl}(C)$, then all points on the line segment connecting x and $\bar{x}$, except possibly $\bar{x}$ belongs to $\text{ri}(C)$.

Proof. Assume $\bar{x} \in C \implies x \in \text{ri}(C)$, so there exists $r > 0$ such that $B(x, r) \cap \text{aff}(C) \subseteq C$. Let $\alpha \in (0, 1)$ and set $x_\alpha = \alpha x + (1 - \alpha)\bar{x}$. We have to prove that $x_\alpha \in \text{ri}(C)$ for all $\alpha \in (0, 1)$.

Consider $B(x_\alpha, \alpha r)$, we have to show that $B(x_\alpha, \alpha r) \cap \text{aff}(C) \subseteq C$. Pick $y_\alpha \in B(x_\alpha, \alpha r) \cap \text{aff}(C)$. Observing that y_α belongs both to $B(x_\alpha, \alpha r)$ and $\text{aff}(C)$, in particular there exists $z_\alpha \in \mathbb{R}^n$, $\|z_\alpha\| \leq \alpha r$ such that $y_\alpha = x_\alpha + z_\alpha$, hence $y_\alpha = \alpha x + (1 - \alpha)\bar{x} + z_\alpha = \alpha \underbrace{\left(x + \frac{z_\alpha}{\alpha}\right)}_{\in B(x, r)} + \underbrace{(1 - \alpha)\bar{x}}_{\in C}$.

Observe that $x + \frac{z_\alpha}{\alpha} \in \text{aff}(C)$. Indeed $x \in \text{aff}(C)$. Consider $z_\alpha = y_\alpha - x_\alpha$, then $x_\alpha \in C \implies x_\alpha \in \text{aff}(C)$, while $y_\alpha \in \text{aff}(C)$. Putting all together, $z_\alpha \in \text{parallel space to } \text{aff}(C)$, so we remain in the affine space, as we are summing a point in the affine space and a point in the parallel subspace. Moreover $x + \frac{z_\alpha}{\alpha} \in B(x, r)$ as $\|\frac{z_\alpha}{\alpha}\| \leq \frac{\alpha r}{\alpha} = r$, hence $x + \frac{z_\alpha}{\alpha} \in C$.

Consider now the case $\bar{x} \in \text{cl}(C) \setminus C$. Then there exists a sequence $\{x_k\}_k$ such that $x_k \rightarrow \bar{x}$ and $x_k \in C$ for all k . Apply then the previous argument to all x_k . Define $x_{k,\alpha} = \alpha x + (1 - \alpha)x_k \implies B(x_{k,\alpha}, r\alpha) \cap \text{aff}(C) \subseteq C$ for all k and $\alpha \in (0, 1)$. Then the sequence $x_{k,\alpha} \rightarrow x_\alpha = \alpha x + (1 - \alpha)\bar{x}$. For k large enough, $B(x_\alpha, \frac{\alpha r}{2}) \subseteq B(x_{k,\alpha}, r\alpha)$. Take the intersection with the affine hull, hence $B(x_\alpha, \frac{\alpha r}{2}) \cap \text{aff}(C) \subseteq B(x_{k,\alpha}, r\alpha) \cap \text{aff}(C)$, showing that $x_\alpha \in \text{ri}(C)$. ■

LECTURE 6 OF 24/03/2026

PROPOSITION 1 (Nonemptiness of relative interiors). *Let $C \subseteq \mathbb{R}^d$ be convex and nonempty. Then:*

- a) *the set $\text{ri}(C)$ is a nonempty convex set, $\text{aff}(\text{ri}(C)) = \text{aff}(C)$;*
- b) *if $m = \dim_{\mathbb{R}}(\text{aff}(C))$, $m > 0$ then there exists $m + 1$ vectors $x_0, x_1, \dots, x_m \in \text{ri}(C)$ such that*

$$\text{aff}(C) = \text{span}\{x_1 - x_0, x_2 - x_0, \dots, x_m - x_0\}.$$

Proof. The fact that $\text{ri}(C)$ is convex is a consequence of LSP. We have to show that $\text{ri}(C)$ is nonempty. Without loss of generality, we can assume $0 \in C$ (up to a traslation), consider $\text{aff}(C)$. If $m = 0$ then $C = \{0\} \implies \text{ri}(C) = C$ as for any $r > 0$, $B(0, r) \cap \text{aff}(C) = C$. Assume now $m > 0$. Consider $z_1, \dots, z_m \in C$ linearly independent such that $\text{aff}(C) = \text{span}\{z_1, \dots, z_m\}$. Consider the set

$$X = \left\{ x : x = \sum_{i=1}^m \alpha_i z_i, \alpha_i > 0, \sum_i \alpha_i < 1 \right\}.$$

The strategy now is to prove that $X \subseteq \text{ri}(C)$. Observe that $X \subseteq C$. Fix $\bar{x} \in X$, $x \in \text{aff}(C)$ and observe that

$$\begin{aligned} \bar{x} &= Z\bar{\alpha}, \bar{\alpha} \in \mathbb{R}^m \\ x &= Z\alpha, \alpha \in \mathbb{R}^m \end{aligned}$$

with Z the matrix with columns $z_1, \dots, z_m$. Notice that

$$\begin{aligned} \|x - \bar{x}\|^2 &= \langle Z(\alpha - \bar{\alpha}), Z(\alpha - \bar{\alpha}) \rangle \\ &= (\alpha - \bar{\alpha})^\top Z^\top Z(\alpha - \bar{\alpha}) \\ &\leq \gamma \|\alpha - \bar{\alpha}\|^2. \end{aligned}$$

Since $\bar{x} \in X$, the vector $\bar{\alpha} \in A = \{\alpha \in \mathbb{R}^m : \sum_i \alpha_i < 1, \alpha_i > 0\}$ which is an open set. If x is sufficiently close to $\bar{x}$, then α is close to $\bar{\alpha}$ by the above estimate on the norm. This implies that $\alpha \in A$, hence $x \in X$, so $B(\bar{x}, r) \cap \text{aff}(C) \subseteq X \subseteq C$. This shows that $X \subseteq \text{ri}(C)$.

Notice that, by construction, $\text{aff}(X) = \text{aff}(C)$, hence $\text{aff}(X) \subseteq \text{aff}(\text{ri}(C)) \subseteq \text{aff}(C)$.

b): fix $x_0 \in \text{ri}(C)$ (x_0 exists by a)), translate C to $C - x_0$, so that $0 \in C - x_0$. fix $z_1, \dots, z_m \in C - x_0$ linearly independent such that $\text{aff}(C - x_0) = \text{span}\{z_1, \dots, z_m\}$. Let $\alpha \in (0, 1)$ and consider $(1 - \alpha) \cdot 0 + \alpha z_i \in \text{ri}(C - x_0)$ by LSP as $0, z_i \in \text{ri}(C - x_0)$. Hence $\alpha z_i \in \text{ri}(C - x_0)$. It follows that the vectors $x_i = x_0 + \alpha z_i$ are such that $x_1 - x_0, \dots, x_m - x_0$ spans the parallel space. ■

LEMMA 2 (Prolongation lemma). Let $C \subseteq \mathbb{R}^d$ convex nonempty. A vector $x \in C$ belongs to $\text{ri}(C)$ if and only if for any $\bar{x} \in C$ there exists a $\gamma > 0$ such that $x + \gamma(x - \bar{x}) \in C$, i.e. every line segment connecting x to any $\bar{x}$ can be prolonged beyond x without leaving C .

Proof. " $\implies$ " easy.

Let x satisfy $(*)$, fix $\bar{x} \in \text{ri}(C)$ (it exists thanks to the above proposition). If $x = \bar{x}$ then $x \in \text{ri}(C)$ and we are happy. If not, there exists a $\gamma > 0$ such that $z = x + \gamma(x - \bar{x}) \in C$ so that x lies within the line segment connecting $\bar{x}$ and z . By LSP, it follows that $x \in \text{ri}(C)$. ■

PROPOSITION 3 (Minima of concave functions). Let $X \subseteq \mathbb{R}^n$ be convex and nonempty. Let $f : X \rightarrow \mathbb{R}$ be concave. Define

$$X^* = \arg \min_{x \in X} f(x).$$

If $X^* \cap \text{ri}(X) \neq \emptyset$, then f is constant.

Proof. Assume $x^* \in X^* \cap \text{ri}(X)$, by prolongation lemma, there exists $\gamma > 0$ such that the point $\hat{x} = x^* + \gamma(x^* - x)$ belongs to X , implying that

$$x^* = \frac{1}{\gamma + 1} \hat{x} + \frac{\gamma}{\gamma + 1} x.$$

By concavity of f ,

$$\begin{aligned}
f(x^*) &= f\left(\frac{1}{\gamma+1}\hat{x} + \frac{\gamma}{\gamma+1}x\right) \\
&\geq \frac{1}{\gamma+1}f(\hat{x}) + \frac{\gamma}{\gamma+1}f(x).
\end{aligned}$$

Since x^* is a minimum point, it must be that $f(y) \geq f(x^*)$ for any y , in particular, $f(\hat{x}) \geq f(x^*)$ and $f(x) \geq f(x^*)$, hence $f(x) = f(x^*)$ for any x . ■

An immediate consequence is that if $f(x) = a^\top x$ with $a \in \mathbb{R}^n$, then the minima of this function is attained at the boundary of its domain (which is assumed to be convex).

PROPOSITION 4 (Continuity of convex functions). *If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, then f is continuous. More generally, if $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ is proper and convex, then the restriction of f to $\text{dom}(f)$ is continuous in $\text{ri}(\text{dom}(f))$.*

Proof. Assume $0 \in \text{dom}(f)$ (if not translate). Without loss of generality, assume that the set $X = \{x : \|x\|_\infty \leq 1\} \subseteq \text{dom}(f)$ (up to dilatation). It is now sufficient to prove that f is continuous at 0: for any sequence $\{x_k\}_k$ such that $kx_k \rightarrow 0$ we want to prove $f(x_k) \rightarrow f(0)$. Define e_i , for $i = 1, \dots, 2^n$ the corners of X . Let $x \in X$, then

$$x = \sum_{i=1}^{2^n} \alpha_i e_i \quad \sum_i \alpha_i = 1, \alpha_i \geq 0.$$

Then

$$\begin{aligned}
f(x) &= f\left(\sum_{i=1}^{2^n} \alpha_i e_i\right) \\
&\leq \sum_i \alpha_i f(e_i) \\
&\leq \sum_i \alpha_i \left(\max_j f(e_j)\right) \\
&\leq \max_j f(e_j).
\end{aligned}$$

Consider now the sequence $x_k \rightarrow 0$, define $y_k = x_k / \|x_k\|$, $z_k = -x_k / \|x_k\|$. We can see x_k as a convex combination of y_k and 0, hence

$$\begin{aligned} f(x_k) &= f(\|x_k\|_\infty y_k - (1 - \|x_k\|_\infty) \cdot 0) \\ &\leq \|x_k\| f(y_k) + (1 - \|x_k\|) f(0). \end{aligned}$$

Observe that $\|x_k\| \rightarrow 0$, and $f(y_k)$ is bounded ($y_k \in \text{dom}(f)$), hence $\|x_k\| f(y_k) \rightarrow 0$ as $k \rightarrow +\infty$. This shows that

$$\limsup_k f(x_k) \leq f(0).$$

Do similarly with z_k :

$$f(0) = f\left(\frac{1}{1 + \|x_k\| x_k} + \frac{\|x_k\|}{1 + \|x_k\| 0}\right) \leq \liminf_k f(x_k).$$

Hence $f(0) \leq \liminf_k f(x_k) \leq \limsup_k f(x_k) \leq f(0)$. This shows that f is continuous at 0. ■

6.1 INTERMEZZO

Let $E \subseteq \mathbb{R}^{n+1}$, $f : D \rightarrow \overline{\mathbb{R}}$ where $D = \{x \in \mathbb{R}^n : \exists w \in \mathbb{R} \text{ s.t. } (x, w) \in E\}$, $f(x) = \inf_w (x, w) \in E$.

Let now $f : X \rightarrow \overline{\mathbb{R}}$ any function. We define the *closure of f* as

$$\text{cl}(f)(x) = \inf \{w : (x, w) \in \text{cl}(\text{epi}(f))\}.$$

The *convex closure* of f is

$$\check{\text{cl}}(f)(x) = \inf \{w : (x, w) \in \text{cl}(\text{conv}(\text{epi}(f)))\}.$$

It can be proven that $\check{\text{cl}}(f) = \text{cl}(F)$ where

$$F(x) = \inf \{w : (x, w) \in \text{conv}(\text{epi}(f))\}.$$

PROPOSITION 1. Let $f : X \rightarrow \overline{\mathbb{R}}$. Then

$$\inf_{x \in X} f(x) = \inf_{x \in \mathbb{R}^n} \text{cl}(f)(x) = \inf_{x \in \mathbb{R}^n} F(x) = \inf_{x \in \mathbb{R}^n} \check{\text{cl}}(f)(x).$$

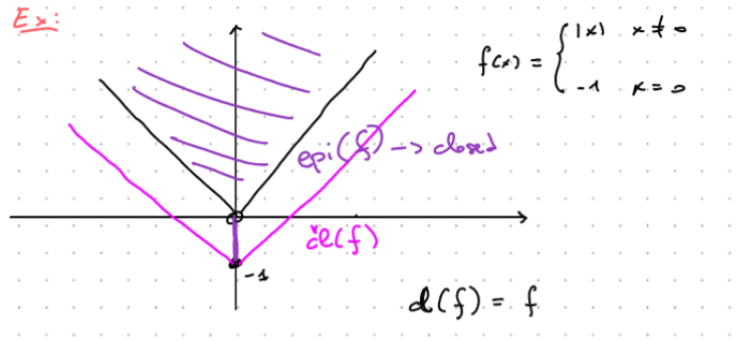


Figure 6.1: Example of convex closure of a function

Proof. If $\text{epi}(f) = \emptyset$, then $f \cong +\infty$, so this case is verified. Suppose now $\text{epi}(f) \neq \emptyset$. Let $f^* = \inf_{x \in \mathbb{R}^n} \text{cl}(f)(x)$. for any sequence $\{(\bar{x}_k, \bar{w}_k)\}_k \subseteq \text{cl}(\text{epi}(f))$ with $\bar{w}_k \rightarrow f^*$ we can construct a subsequence $\{(x_k, w_k)\}_k \subseteq \text{epi}(f)$ such that $w_k \rightarrow f^*$, $|w_k - \bar{w}_k| \rightarrow 0$. For each $x_k \in X$, it holds $(x_k, w_k) \in \text{epi}(f)$, then $f(x_k) \leq w_k$. But then

$$\inf_{x \in X} f(x) \leq \limsup_k f(x_k) \leq \lim_k w_k = f^* \leq \text{cl}(f)(x) \leq f(x).$$

Hence

$$\inf_{x \in X} f(x) = \inf_{x \in \mathbb{R}^n} \text{cl}(f)(x).$$

The next part uses the same argument. ■

PROPOSITION 2. Let $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ a function. Then

- a) $\text{cl}(f)$ is the greatest closed function below f (recall that we say $g \leq f$ if $g(x) \leq f(x)$ for any x);
- b) $\check{\text{cl}}(f)$ is the greatest convex closed function below f .

Proof. Let $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ closed below f , then $\text{epi}(f) \subseteq \text{epi}(g)$, but $\text{epi}(\text{cl}(f)) = \text{cl}(\text{epi}(f)) = \text{intersection of all closed sets containing } \text{epi}(f)$. Notice that $\text{epi}(g)$ is closed, as g is closed. Hence $\text{cl}(\text{epi}(f)) \subseteq \text{epi}(g)$ as $\text{epi}(g)$ is one of the closed sets containing $\text{epi}(f)$. This shows that $\text{cl}(f) \leq g$. The same argument works for b). ■

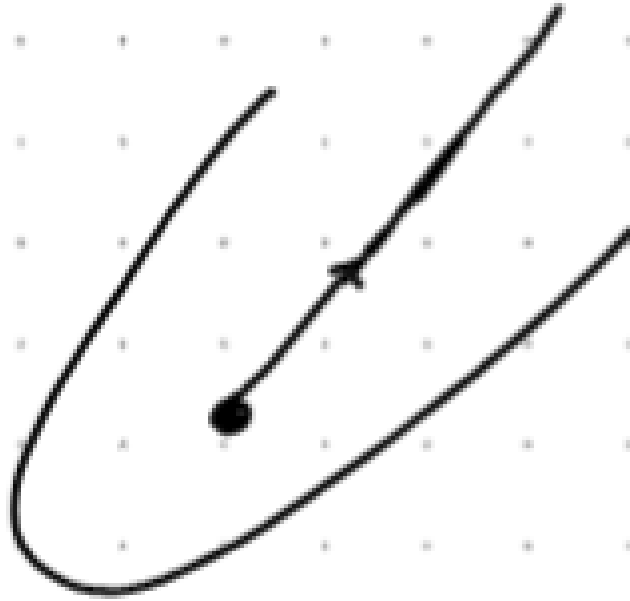


Figure 6.2: Example of recession cone

6.2 RECESSON CONES

DEFINITION 1 (Recession direction/cone). Let C be convex and nonempty. We say that a direction $d \in \mathbb{R}^n$ is a *recession direction* for C if $x + \alpha d \in C$ for any $x \in C, \alpha \geq 0$.

We denote with R_C the *recession cone* of C the set of all recession directions.

Example 2.

$$C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}. \quad R_C = \{0\}. \quad (6.1)$$

$$C = (-\infty, 2]. \quad R_C = (-\infty, 0]. \quad (6.2)$$

PROPOSITION 3 (Recession cone theorem). Let C closed convex nonempty. Then d functions. This methodology is fundamental in several convex optimization contexts, including the issue of existence of optimal solutions, which will be discus

a) R_C is closed and convex;

b) $d \in R_C \iff$ there exists $x \in C$ such that $x + \alpha d \in C$ for all $\alpha \geq 0$.

Proof. We first prove convexity. Let $d_1, d_2 \in R_C, \gamma_1, \gamma_2 \in [0, 1]$ such that $\gamma_1 + \gamma_2 = 1$. Fix any $x \in C, \alpha \geq 0$ and consider $x + \alpha(\gamma_1 d_1 + \gamma_2 d_2) = x(\gamma_1 + \gamma_2) +$

6.2 Recession cones

$\alpha(\gamma_1 d_1 + \gamma_2 d_2) = \gamma_1(x + \alpha d_1) + \gamma_2(x + \alpha d_2)$ which is a convex combination of points in C . For the closeness, consider $\{d_k\}_k$ a sequence in R_C such that $d_k \rightarrow d \in \mathbb{R}^n$. Then for any $x \in C$ and for any $\alpha \geq 0$, $x + \alpha d_k \in C \rightarrow x + \alpha d$ as $k \rightarrow +\infty$. This shows that $x + \alpha d \in C$ as C is closed.

For point b), the implication " $\implies$ " is obvious.

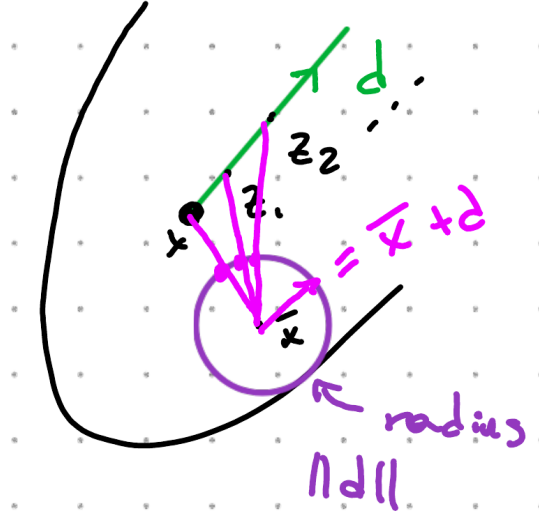


Figure 6.3: Visual intuition on the sequence z_k

Let $d \in \mathbb{R}^n$ and assume $x \in C$ such that $x + \alpha d \in C$ for any $\alpha \geq 0$. Let $\bar{x} \in C$. We have to prove $\bar{x} + \alpha d \in C$. Assume, without loss of generality, $\alpha = 1$ as we can retrieve the general case with a scaling argument. Define a sequence $z_k = x + kd$, $z_k \in C$ for all k . If $\bar{x} = z_k$ for some k then we are done. Assume then $\bar{x} \neq x_k$ and define $d_k = \frac{z_k - \bar{x}}{\|z_k - \bar{x}\|} \|d\|$. Then

$$\begin{aligned} d_k &= \frac{z_k - \bar{x} + x - x}{\|z_k - \bar{x}\|} \|d\| = \frac{\overbrace{z_k - x}^{=kd}}{\|z_k - \bar{x}\|} \|d\| + \frac{x - \bar{x}}{\|z_k - \bar{x}\|} \\ &= \frac{k\|d\|}{\|z_k - \bar{x}\|} d + \frac{x - \bar{x}}{\|z_k - \bar{x}\|}. \end{aligned}$$

The last term goes to 0, indeed, $x - \bar{x}$ is fixed, $\|d\|$ is fixed and $\|z_k\| \nearrow +\infty$ as $k \nearrow +\infty$. I claim that the first term goes to d : in fact,

$$\frac{k\|d\|}{\|x + kd - \bar{x}\|} = \frac{k\|d\|}{k\|d + (x - \bar{x})/k\|} \rightarrow 1.$$

This shows that $d_k \rightarrow d$ as $k \nearrow +\infty$. For k large enough, the vector $\bar{x} + dk$ belongs to the segment connecting $\bar{x}$ to z_k , so, by convexity of C , $\bar{x} + d_k \in C$, hence $\bar{x} + d \in C$ as C is closed. ■

PROPOSITION 4 (Properties of the recession cone). *Let $C \subseteq \mathbb{R}^n$ closed convex nonempty. Then*

- a) R_C contains a nonzero direction if and only if C is unbounded;
- b) $R_C = R_{\text{ri}(C)}$
- c) if $\{C_i\}_{i \in I}$ is a collection of closed convex sets such that $\bigcap_{i \in I} C_i \neq \emptyset$, then $R_{\bigcap_{i \in I} C_i} = \bigcap_{i \in I} R_{C_i}$;
- d) fix $W \subseteq \mathbb{R}^n$ compact, $A \in \mathbb{R}^{n \times n}$ a matrix. Let $V = \{x \in C : Ax \in W\}$. Assume V nonempty. Then $R_V = R_C \cap N(A)$ where $N(A) = \{x \in \mathbb{R}^n : Ax = 0\}$ is the null space of the linear operator A .

Proof. Not required. ■

Example 5. The following example shows how the closeness of C can play a role in the recession cone. Consider the set

$$C = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, y \geq 0\} \cup \{(1, 0)\}.$$

Then $R_C = \{0\}$, indeed, if we start from the point $(1, 0)$, we cannot move along any direction without leaving C , while $R_{\text{ri}(C)} = \text{span}\{(0, 1)\}$.

6.3 LINEARITY SPACE

We define the linearity space for a set C as the set

$$L_C = R_C \cap (-R_C).$$

In other words, $d \in L_C$ if and only if $x + \alpha d \in C$ and $x - \alpha d$ are inside C for any $\alpha \geq 0$.

PROPOSITION 1 (Properties of L_C). *Let $C \subseteq \mathbb{R}^n$ convex closed nonempty, then*

- 1. L_C is a subspace;
- 2. $L_C = L_{\text{ri}(C)}$;

3. same as above.

Proof. Not required. ■

Example 2 (Important!). Consider the set

$$C = \left\{ x \in \mathbb{R}^n : x^\top Qx + c^\top x + b \leq 0 \right\}$$

where $Q \in \mathbb{R}^{n \times n}$ is SPD, $c \in \mathbb{R}^n$, $b \in \mathbb{R}$. We want to compute R_C .
Let $x \in C$, $d \in \mathbb{R}^n$. Then $x + \alpha d \in C$ for any $\alpha \geq 0$ if and only if

$$\begin{aligned} (x + \alpha d)^\top Q(x + \alpha d) + c^\top (x + \alpha d) + b &\leq 0 \\ \iff \underbrace{x^\top Qx + c^\top x + b}_{\leq 0} + \alpha(c^\top d + 2x^\top Qd) + \alpha^2 d^\top Qd &\leq 0 \end{aligned}$$

for any $\alpha \geq 0$ and for any $x \in C$.

The quantity $\alpha^2 d^\top Qd \geq 0$ since Q is SPD, hence we need $d^\top Qd = 0$. Since Q is SPD, we can compute its Cholesky factorization, hence $Q = M^\top M \implies d^\top M^\top M d = 0$, so $Md = 0$.

Furthermore, we need also $\alpha c^\top d \leq 0$. Finally, the recession cone turns out to be

$$R_C = \left\{ d \in \mathbb{R}^n : Qd = 0, c^\top d \leq 0 \right\}.$$

Moreover,

$$L_C = \left\{ d \in \mathbb{R}^n : Qd = 0, c^\top d = 0 \right\}.$$

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PROPOSITION 1. Let $C \subseteq \mathbb{R}^n$ nonempty convex, let $S \subseteq L_C$ be a subspace. Then we can decompose C as

$$C = S + (C \cap S^\perp).$$

Proof. Not required. ■

PROPOSITION 2. Let $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ be closed proper convex function. Let $V_\gamma = \{x \in \mathbb{R}^n : f(x) \leq \gamma\}$ for all $\gamma \in \mathbb{R}$ (sublevel sets of f). Then

a) all nonempty V_γ have the same recession cone, denoted by

$$R_f = \left\{ d \in \mathbb{R}^n : (d, 0) \in R_{\text{epi}(f)} \right\}$$

b) if one V_γ is compact, then all the sublevel sets are compact.

Proof. Fix $\gamma \in \mathbb{R}$, assume $V_\gamma \neq \emptyset$. Define the γ -slice $S = \{(x, \gamma) : f(x) \leq \gamma\}$. Observe that $S = \text{epi}(f) \cap \{(x, \gamma) : x \in \mathbb{R}^n\}$. We already know $\text{epi}(f)$ closed convex, so

$$R_S = R_{\text{epi}(f)} \cap R_{\{(x, \gamma) : x \in \mathbb{R}^n\}} = R_{\text{epi}(f)} \cap \{(d, 0) : d \in \mathbb{R}^n\} = \left\{ (d, 0) : d \in R_{\text{epi}(f)} \right\}$$

which does not depend on γ .

Point b): from a), we know $R_{V_{\gamma_1}} = R_{V_{\gamma_2}}$ for any $\gamma_1, \gamma_2 \in \mathbb{R}$ such that their sublevel sets are nonempty. If, for $\bar{\gamma} \in \mathbb{R}$, $V_{\bar{\gamma}}$ is compact, then $R_{V_{\bar{\gamma}}} = \{0\}$. Then for a) every recession cone is equal, hence $R_{V_\gamma} = \{0\}$ for any γ . This shows that V_γ is compact for every $\gamma \in \mathbb{R}$. ■

Remark 3. Let $d \in R_f$ if $d \in R_{V_\gamma}$ for any γ . Fix $\gamma \in \mathbb{R}$, then $d \in R_{V_\gamma}$ implies that for $x \in V_\gamma$, $x + \alpha d \in V_\gamma$ for any $\alpha \geq 0$, hence $f(x + \alpha d) \leq \gamma$ for every $\alpha \geq 0$. This shows that on the direction d , the function f cannot increase, so d is a direction of continuous non-ascent for f .

DEFINITION 4 (Linearity space of a function). The linearity space of a function is defined in the same way as the linearity space of a set.

Remark 5. If $d \in L_f$, then

$$\begin{aligned} f(x + \alpha d) &\leq f(x) \quad \forall \alpha \geq 0 \\ f(x - \alpha d) &\leq f(x) \quad \forall \alpha \geq 0. \end{aligned}$$

I claim that d is a constancy direction for f . Let $x = \frac{1}{2}(x - \alpha d) + \frac{1}{2}(x + \alpha d)$. Indeed, assume by contraddiction $f(x - \alpha d) < f(x)$. Then

$$\begin{aligned} f(x) &= f\left(\frac{1}{2}(x - \alpha d) + \frac{1}{2}(x + \alpha d)\right) \\ &\leq \frac{1}{2} \underbrace{f(x - \alpha d)}_{< f(x)} + \frac{1}{2} \underbrace{f(x + \alpha d)}_{\leq f(x)} < f(x). \end{aligned}$$

which is a contraddiction.

This shows that $f(x + \alpha d) = f(x - \alpha d) = f(x)$ for any $\alpha \geq 0$. The space L_f is also called *constancy space*.

Example 6. Consider $f(x) = x^\top Qx + c^\top x + b$ as before. It is not difficult to show that

$$L_f = \left\{ d : Qd = 0, c^\top d = 0 \right\}.$$

7.1 RECESSION FUNCTION

Consider $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ closed proper convex. This implies $\text{epi}(f)$ is closed convex subset of $\mathbb{R}^{n+1}$. Consider $R_{\text{epi}(f)} \subseteq \mathbb{R}^{n+1}$. Question: is that set the epigraph of something? The answer is yes, for every x , the quantity $\inf_w \left\{ (d, w) : d \in R_{\text{epi}(f)} \right\}$ is finite or $+\infty$. Since $(d, w) \in R_{\text{epi}(f)}$ then $(d, f(x)) + \alpha(d, w) \in \text{epi}(f)$, hence $(x + \alpha d, f(x) + \alpha w) \in \text{epi}(f) \implies f(x + \alpha d) \leq f(x) + \alpha w$. Assume w can be taken as negative as we want, so $f(x + d) = -\infty$, but this is not possible. This implies that $R_{\text{epi}(f)}$ can be the epigraph of some function.

DEFINITION 1 (Recession function). We define r_f the recession function of f as the function such that $\text{epi}(r_f) = R_{\text{epi}(f)}$.